IT462 - EDA

Tools - An open MIND
Closed HEART

- Python - (Google Colab)

numpy
Pandas

17, 41, 30, 99

Data

Scores with AVG

int 50

Data

raw / unorganized unstructured

information

Text

audio

Visuals

Records

Structured

Source

Primary Data (I am involved)

Secondary Data (You are involved)

Tertiary data

Data

File formats

- .csv
- .xlsx

Text

- .json
- .html

Webscraping

Python

Data

Statistics

tools package

Usable

IDA

Initial Data Analysis

- Validity
- reliability
- authenticity
- error free

ID	IT462	Age	Gender
2026.	10	0	M
2026	50	10	F
. . .	90	20	M
	20	15	F
2023	17	:	F

IDA

EDA

CDA

Statistical aspects

Problem

Data collection

Preprocessing cleaning

features of Data

variables

Modelling
Evaluation

Structured Data

Columns → variables or features

rows → observations or values

- categorical (Qualitative)
- Numerical (Quantitative)

Measurement Scales

- Categorical
  - Nominal
  - Ordinal

Nominal — Names / Indexes / Tags

Pokeste
- color — Nominal
- size — Ordinal

Numerical

Interval

Ratio

Nominal

Measurement Scales

1. Nominal
2. Ordinal
3. Interval
4. Ratio



Category → Labels / Category

Numeric data → quantity

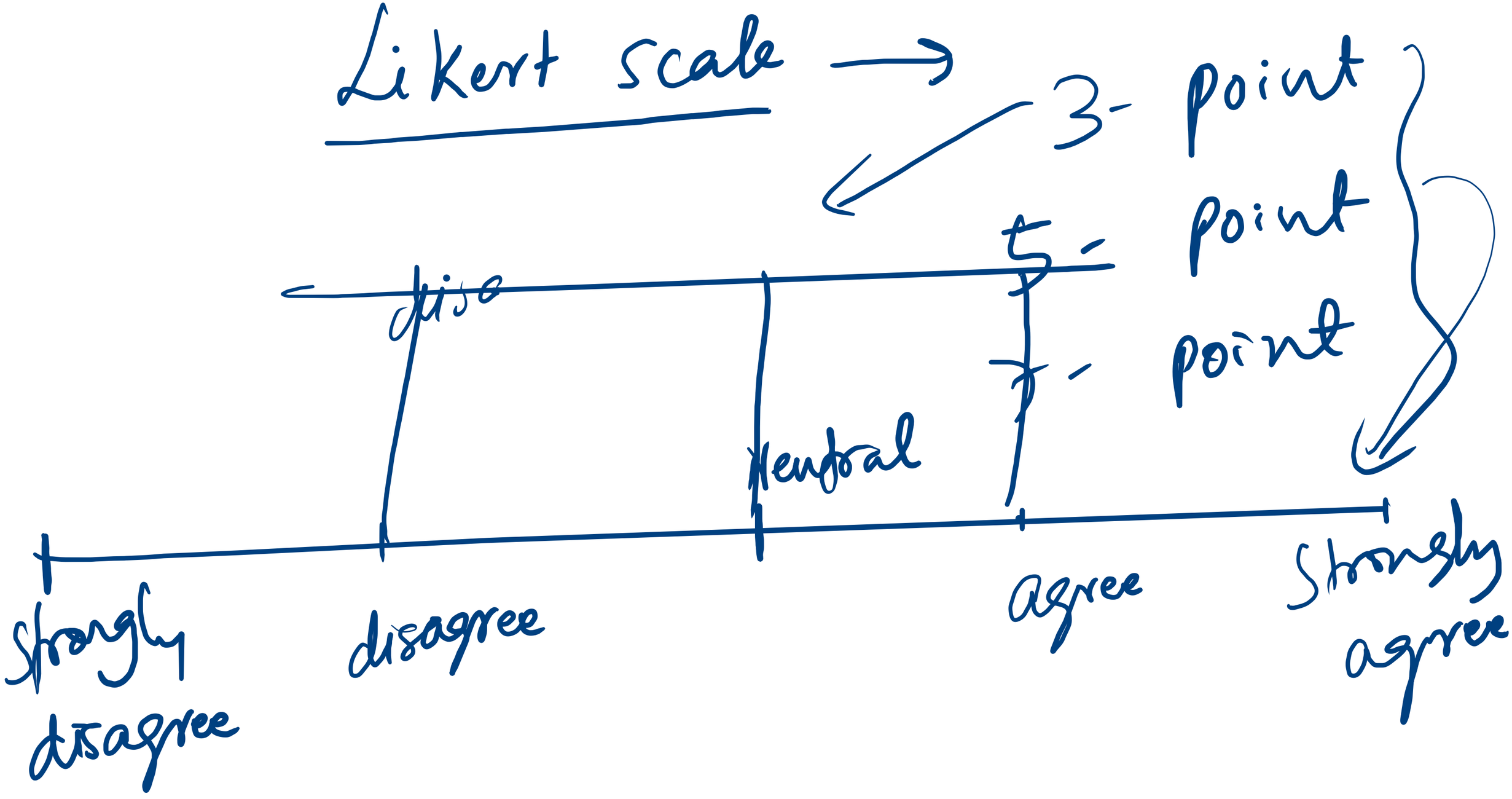
Data	N	O	I	R
Labeled Data	✓	✓	✓	✓
inherent order	X	✓	✓	✓
Measurable difference	X	X	✓	✓
Absolute Zero (True Zero)	X	X	X	✓

Likert scale →

3- point

5- point

7- point



Ordinal → transform → Likert Scale → Interval

- make quantifiable equivalence with Numbers
- make the difference equal in Length



Age

Ratio (because of Absolute Zero Person wh age 0 is impossible)

Variables | features

values

Operations	N	O	I	R
	X	X	$+$ $-$	X $\frac{0}{0}$

Dataset variables

measurement scales

Step 1

Can I count / Identify / Group?

N → X

Y → Nominal

Yes → Rank?

Yes → Ordinal

No

%, freq, Counts

drop

rank!

Step 2

Can I measure the difference;

+, −

Yes → Interval

No

Step 3

Can I find the ratio?

Ratio

×, ÷

Yes

Step 4

Statistical tools

Descriptive

Summary / Description → Data

5 different measures → Central tendency

- mean
- median
- Mode

Inferential

Statistical measures for ↗ measurement scales (completed)

(1) Measures of central tendency

- Mean
  - Arithmetic
  - Geom ( )
  - Harmonic mean
- Median
- Mode

	sort	expression	variants
Mean	×	$\frac{\sum x_i}{n}$	Trimmed — find mean after removing the extremes: x_i, $i \geq 1 \cdots n$ weighted: $\frac{w_i x_i}{\sum w_i}$
Median	✓	↑ n-odd: $x_{(\frac{n+1}{2})}$ n-even: $\frac{1}{2}\left(x_{\frac{n}{2}} + x_{\frac{n}{2}+1}\right)$ median	
Mode	✓		

$i = 1, 2, 4, 6$

② Measures of Dispersion / variation

$x_1 \ldots x_n$

Range :— $x_H - x_L$

x_H - max^m Score

x_L - min^m Score

Variance :— Pop^n $\dfrac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N}$ → mean

Standard deviation :—

Sample :

$\dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$ ✓

$\sqrt{\text{variance}}$

Quartile deviation

Percentiles

Quartile

Quantile

decile :-

Percentile position of x_i

$$P(x_i) = \left(\frac{(i-1)}{(n-1)} \times 100\right)$$

$$\sum \frac{i}{n} \times 100$$

2, 3, 5, 6, 8, 10, 12, 15, 18, 20

1 2 3 4 5 6 7

$n = 10$

$x_i = 12$

What is $P(x_i) = \frac{i-1}{n-1} \times 100 = \frac{7-1}{10-1} \times 100 = \frac{6}{9} \times 100 = 67\%$?

Q1

Q2: Find the 25th percentile ?

$P_{25} = P \times (n-1) = \frac{25 \times 9}{100} = \frac{9}{4} = 2.25$

$i-1 = \frac{P(x_i) \times (n-1)}{100} = 2.25$

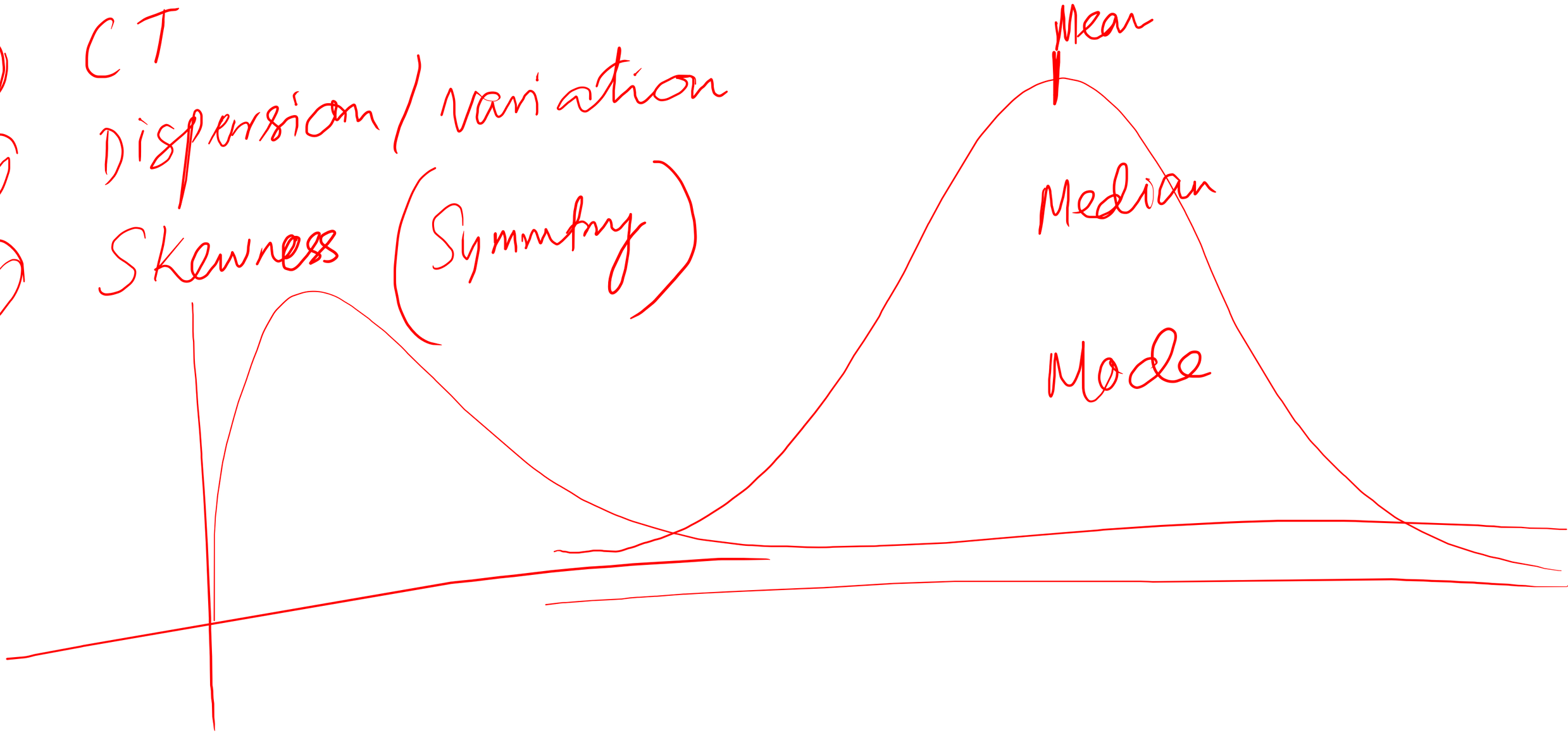
$i = 1 = 2.25$

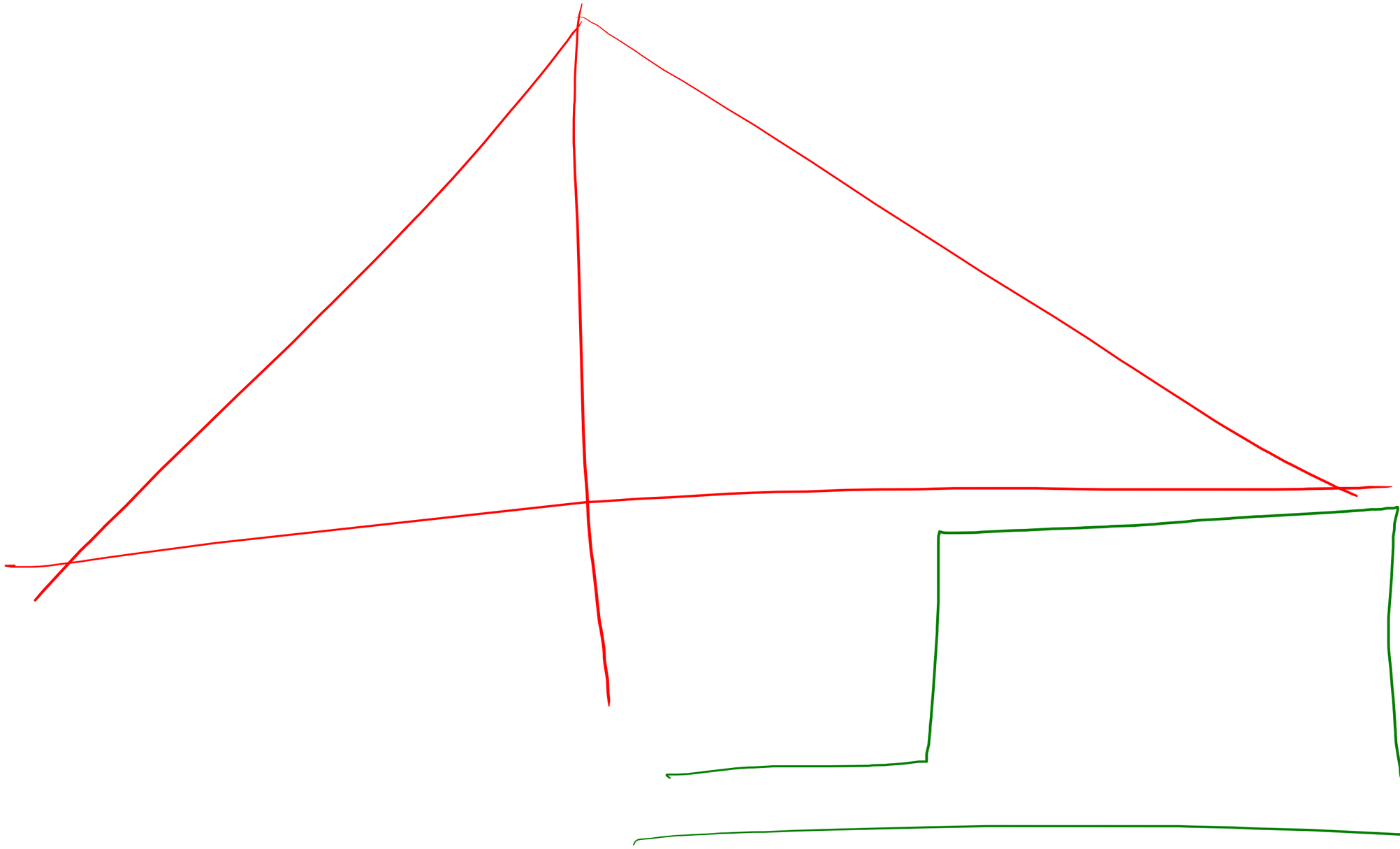
$i = 3.25$

$i-1$

Statistical measures

1. CT
2. Dispersion / variation
3. Skewness (Symmetry)





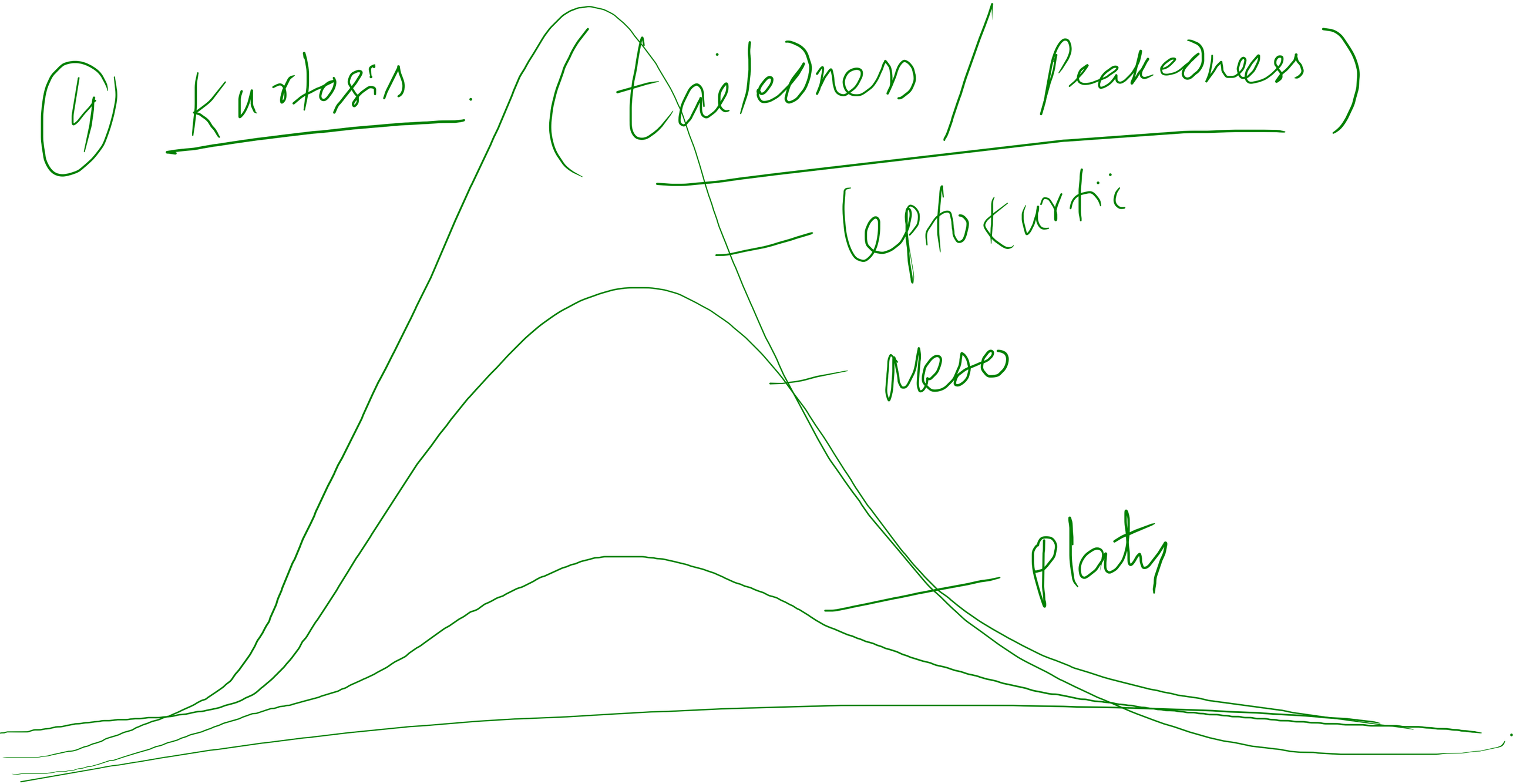
Skew function — Skewness

df

Pandas

df[' '].skew()

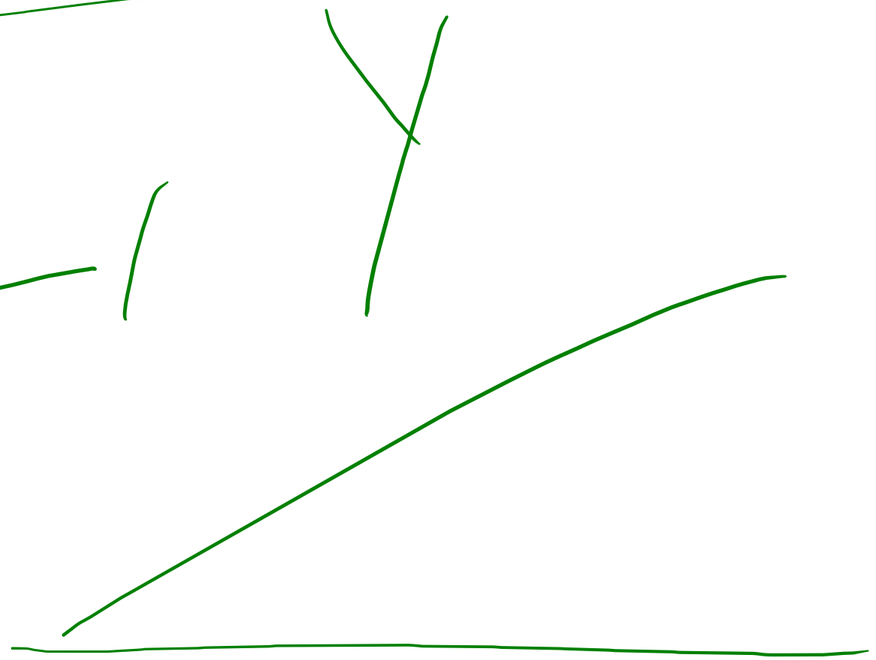
(4) Kurtosis (tailedness / peakedness)



(5) Measures of relationship / association

variables → X , Y

Linear relationship

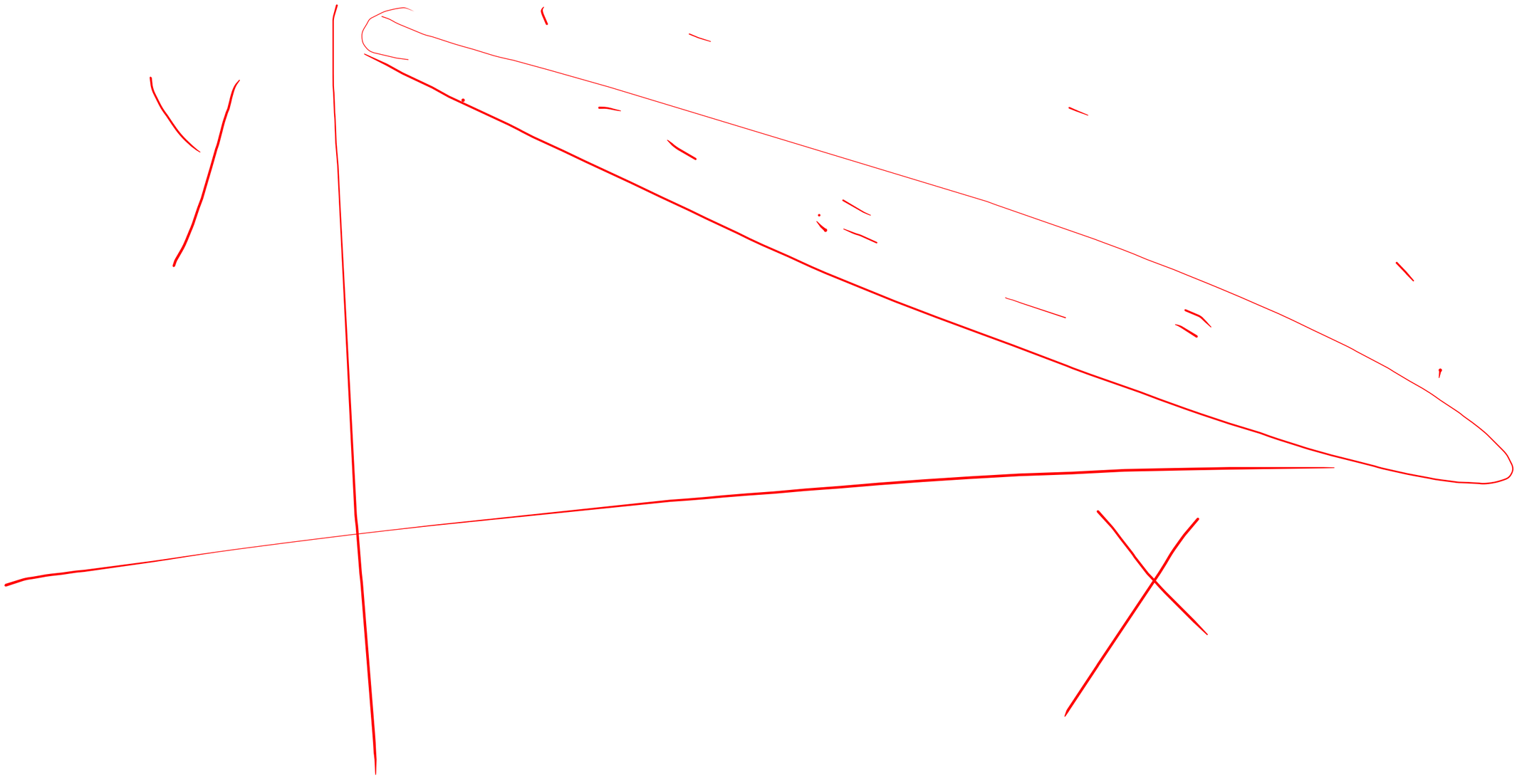


Non-linear relationship

Y

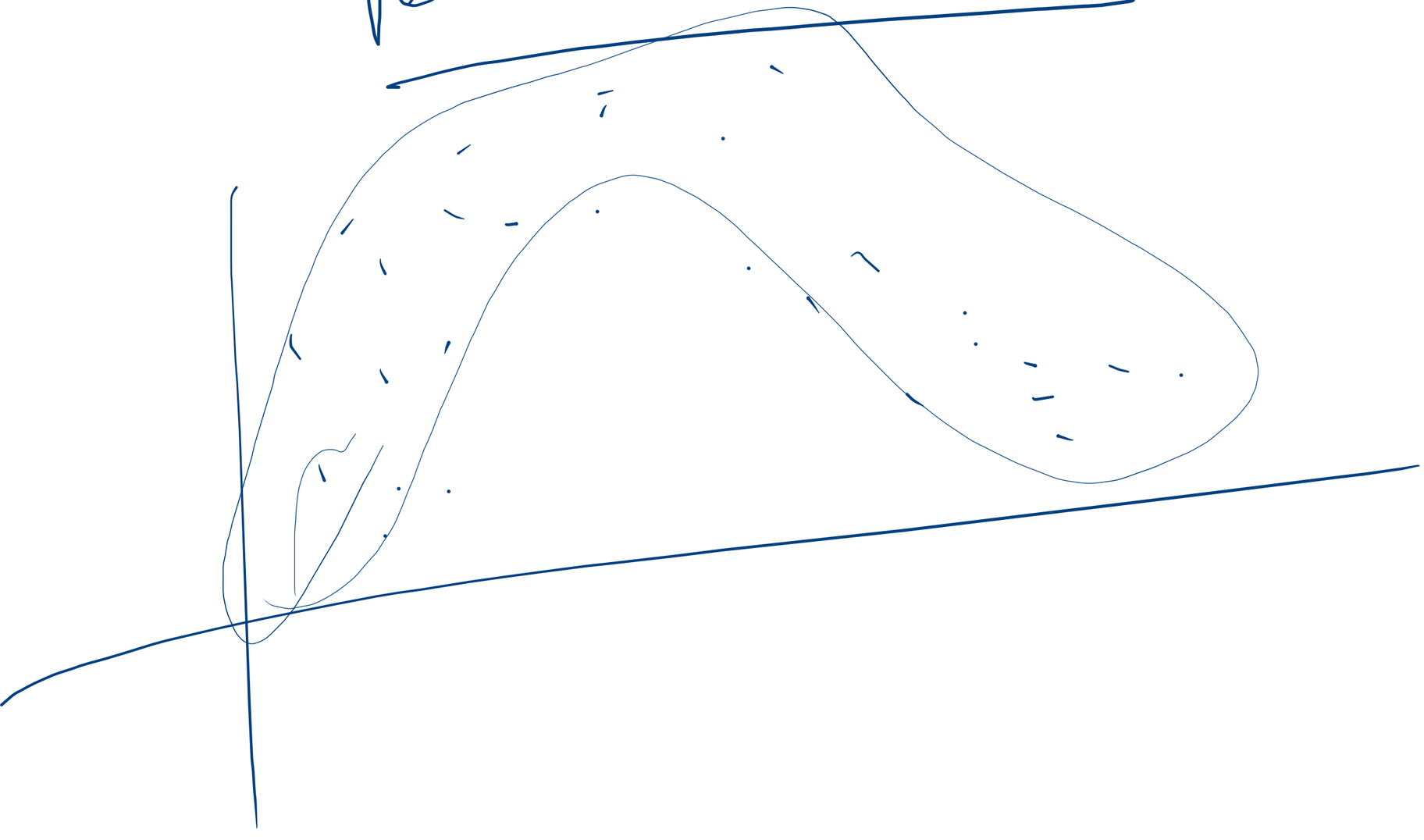
Scatter plot

X



Pearson Correlation

df.corr()



Interval
Ratio
→ Pearson γ

Category → Nominal
→ ordinal
→ X
→ Spearman γ_s

Y - cts
X - Binomial
→ point biserial Correlation

Binomial
Binomial
→ Cramer φ

scipy.stats — ss

Spearman r → ss.spearmanr(x, y)

pointbiserial → ss.pointbiserialr(x, y)

Cramer's φ / V → ——— Crosstab

Contingency table in statsmodels

target

Data ⟶ Statistical tools

↓

characterize (measure scales)

Relation population

Expt

eg Effect of Vaccination

↑

Sampling

n size

N size

population

parameter

sample

each unit/entity observed.

statistic

$N \gg n$

Who should I observe?

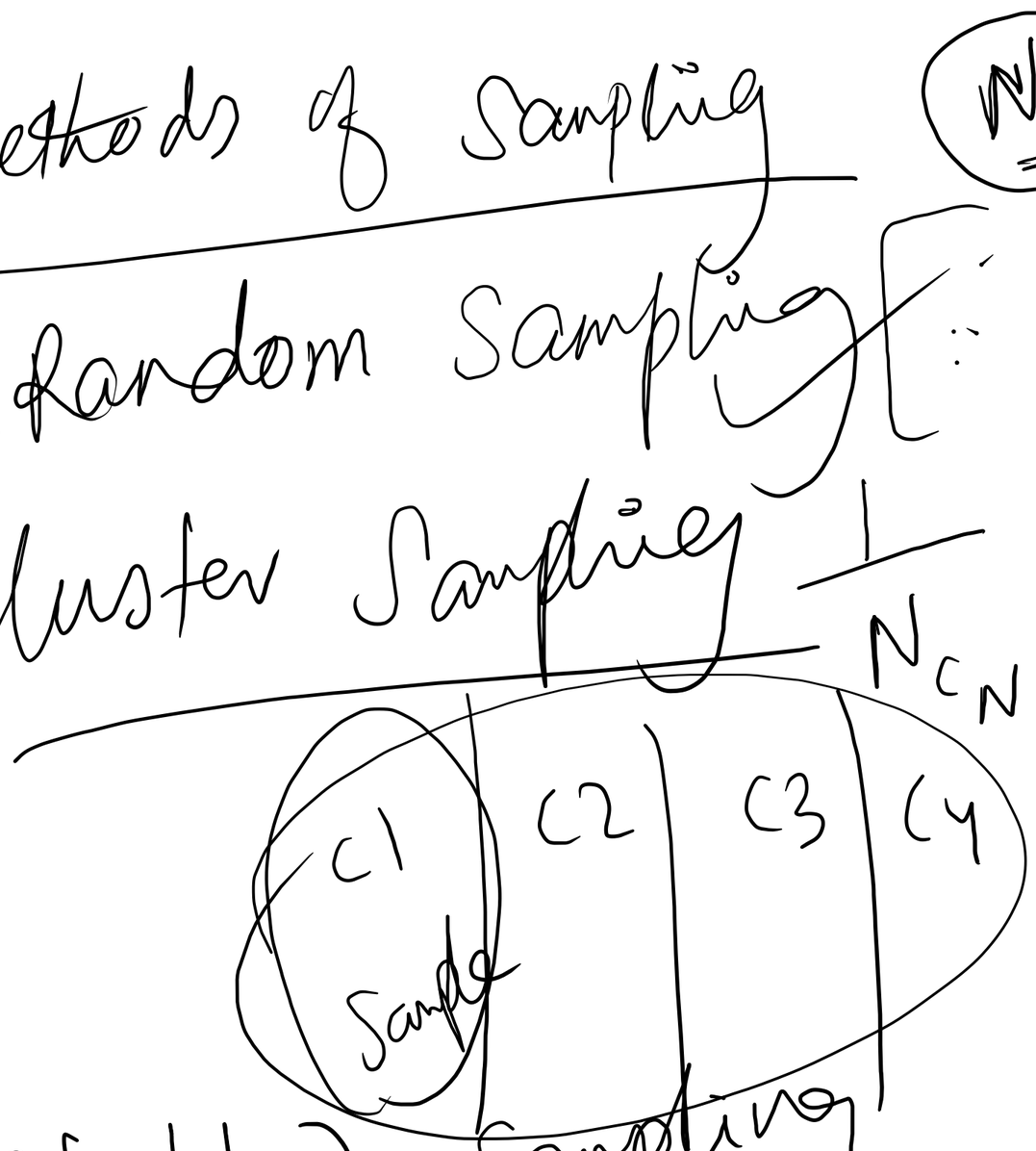
$_{N}C_{n}$

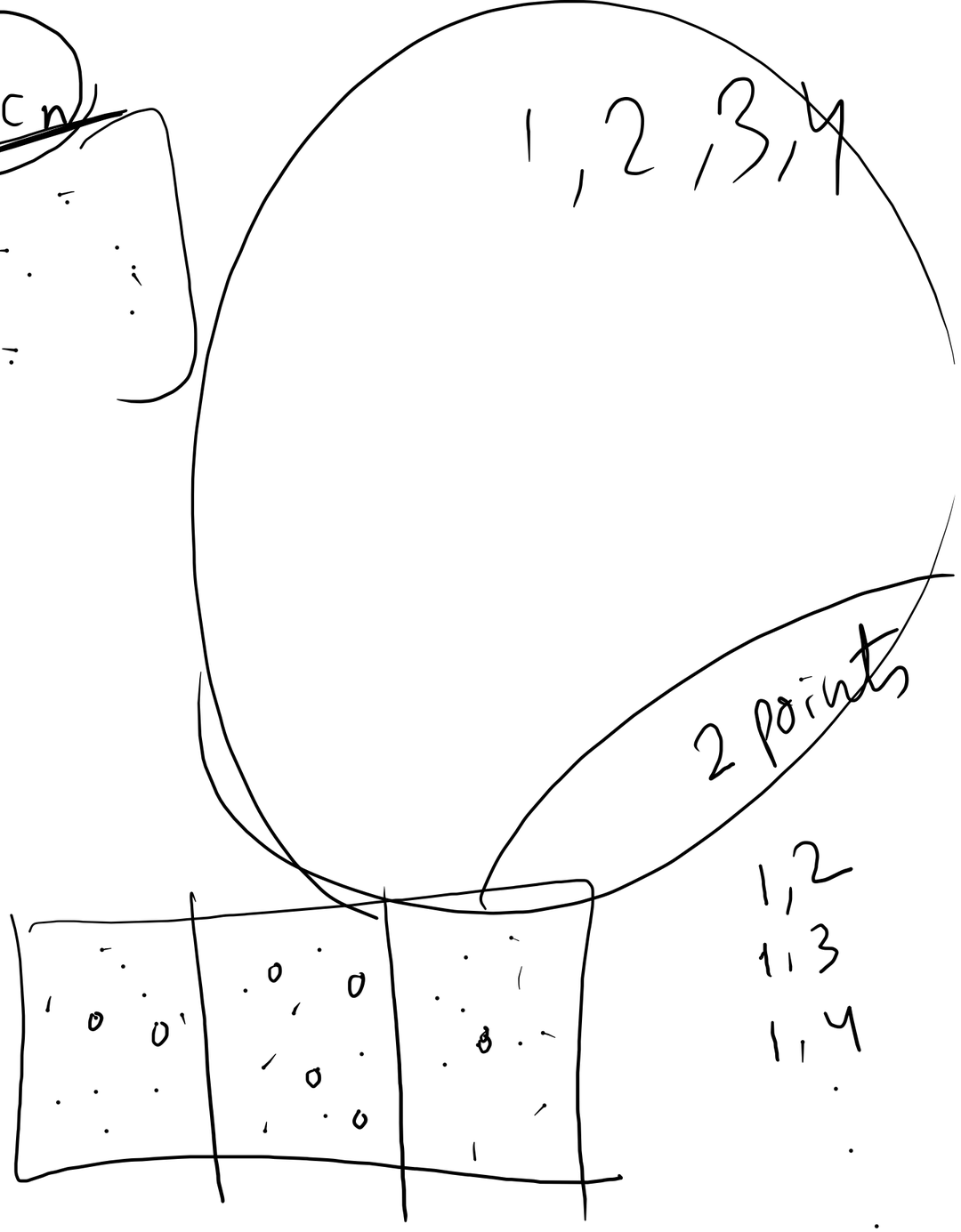
methods of Sampling

- Simple Random Sampling

- Cluster Sampling

- Stratified Sampling





Expt Design

Sampling Bias

Design of Expts

Problem Defn / Research Question

target

Response var

Factors / levels

Experimental Unit

Choose Design

Randomization

Replication

control / Block

Collect data

IDA

EDA

What patterns are there?

Is the Data reliable?

Data Import

(.csv)

① features



① dimension — df.shape

② type (measurement scales)

③ naming convention

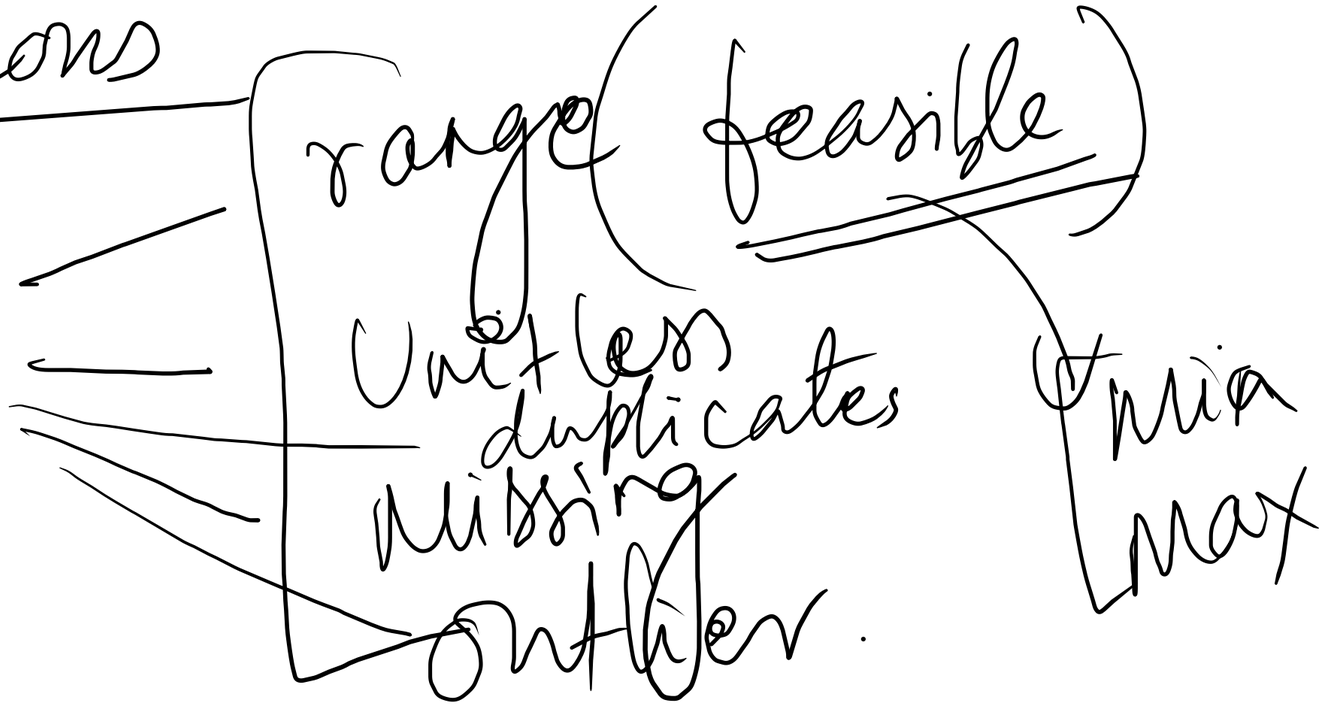
④ Units, date-time (values)

⑤ duplication

⑥ .....

⑦ feature Engineering — add on

② Values / observations



- Imported Data — CSV
- features / columns / variables
  - Nominal
  - ordinal
  - binomial
  - } cat
  - Numeric
- values / observations / records
  - Ranges
  - feasibility
  - duplication
  - outliers
  - Missing values
  - } ?

Definition

Outliers ⇒ values / obs that lie distant from most of the obs

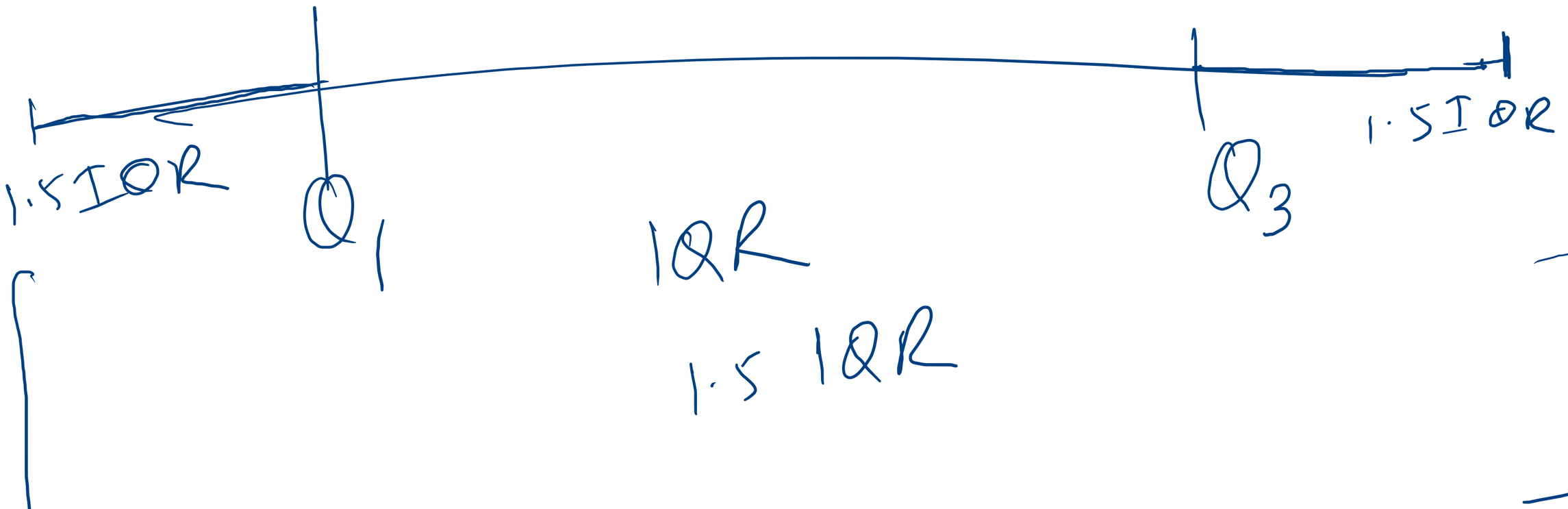
→ extreme values

Detection

Numerical → IQR

$x \notin [Q_1 - 1.5\,IQR,\ Q_3 + 1.5\,IQR]$

Graphical

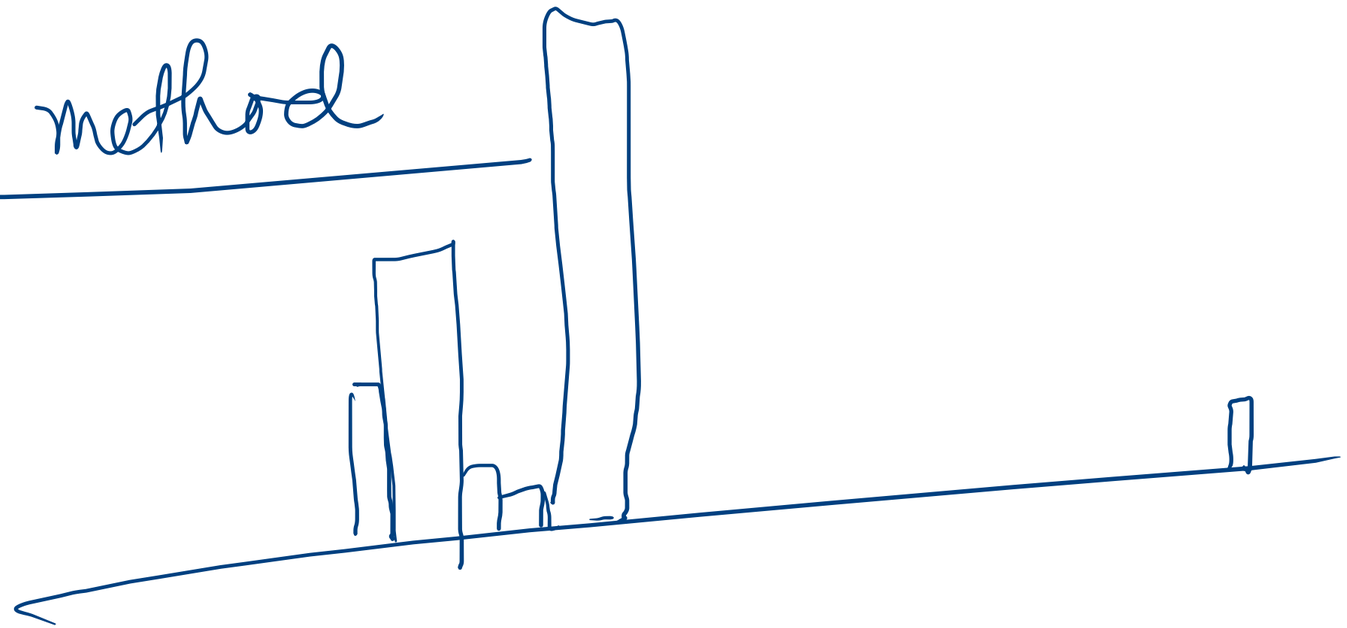


② Z-score

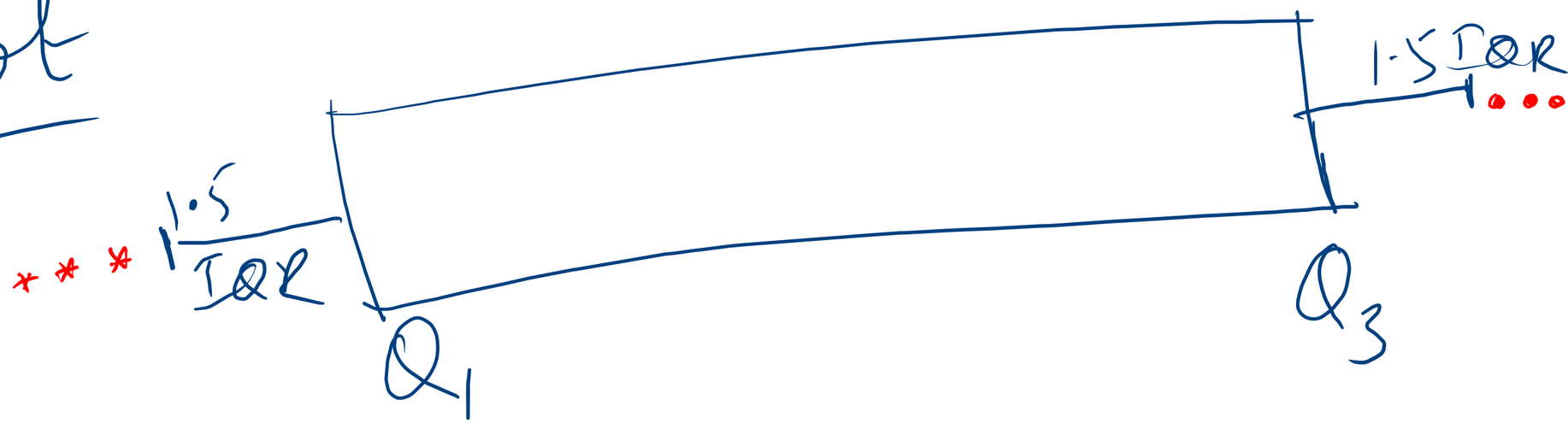
$X \notin \pm 3Z$

Graphical method

① Histogram



② Box plot



Outliers (Univariate) (data points)

(1) Definition → rare or extreme values

(2) Detection — Numerical — Z-score
IQR = $Q_3 - Q_1$

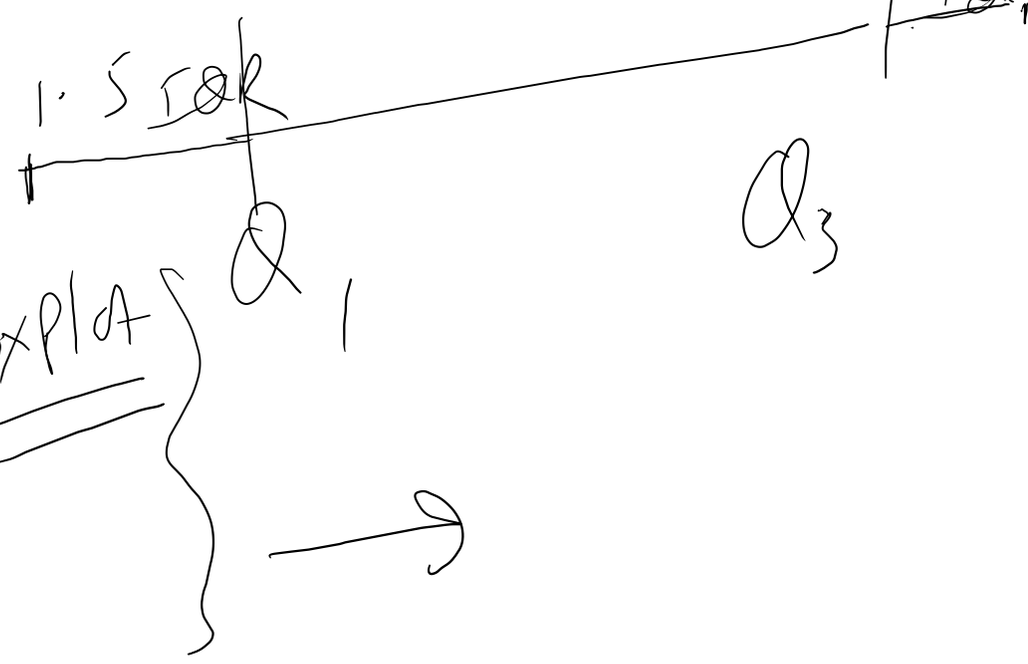
Graphical

(3) Treatment —

Histogram

PDF
≡ CDF

BoxPlot



Multivariate

IQR

extend it to

Correlation matrix

Complex / high dimension data

Graphical

- Heatmap
- Scatter plots
- Pair plots

- PCA
- LOF
- PyOD
- AOD

Treatment

- Remove
- Replace → Range random, Median, Mean, Percentile — Numeric data
- Imputation → Use Specific ML algorithms

Winsorization

Trimming

Missing Data Mechanism (MDM)

Missing values analysis

Defⁿ :- Absence of data values ← absence inconsistent diff. forms

Causes :- - may n't exist / error -

Classification :- → Rubin (1976)

Detection :

Treatment :

classificatia

MCAR - Missing Completely At Random

MAR - Missing At Random

NMAR/MNAR - Not " " "

MCAR:

Age Gender Score Y_0 Y_m

MAR

Y_m DEP Y_0 X

IND

Y_m IND Y_0 Other var

Missing Data Mechanism

Defn. —

Cause —

Types :

MCAR / MAR / MNAR

Missing values doesn't depend on the observe values or missing values

Missing values depends on the obs. values

Missing values depend on the missing values itself

Y X

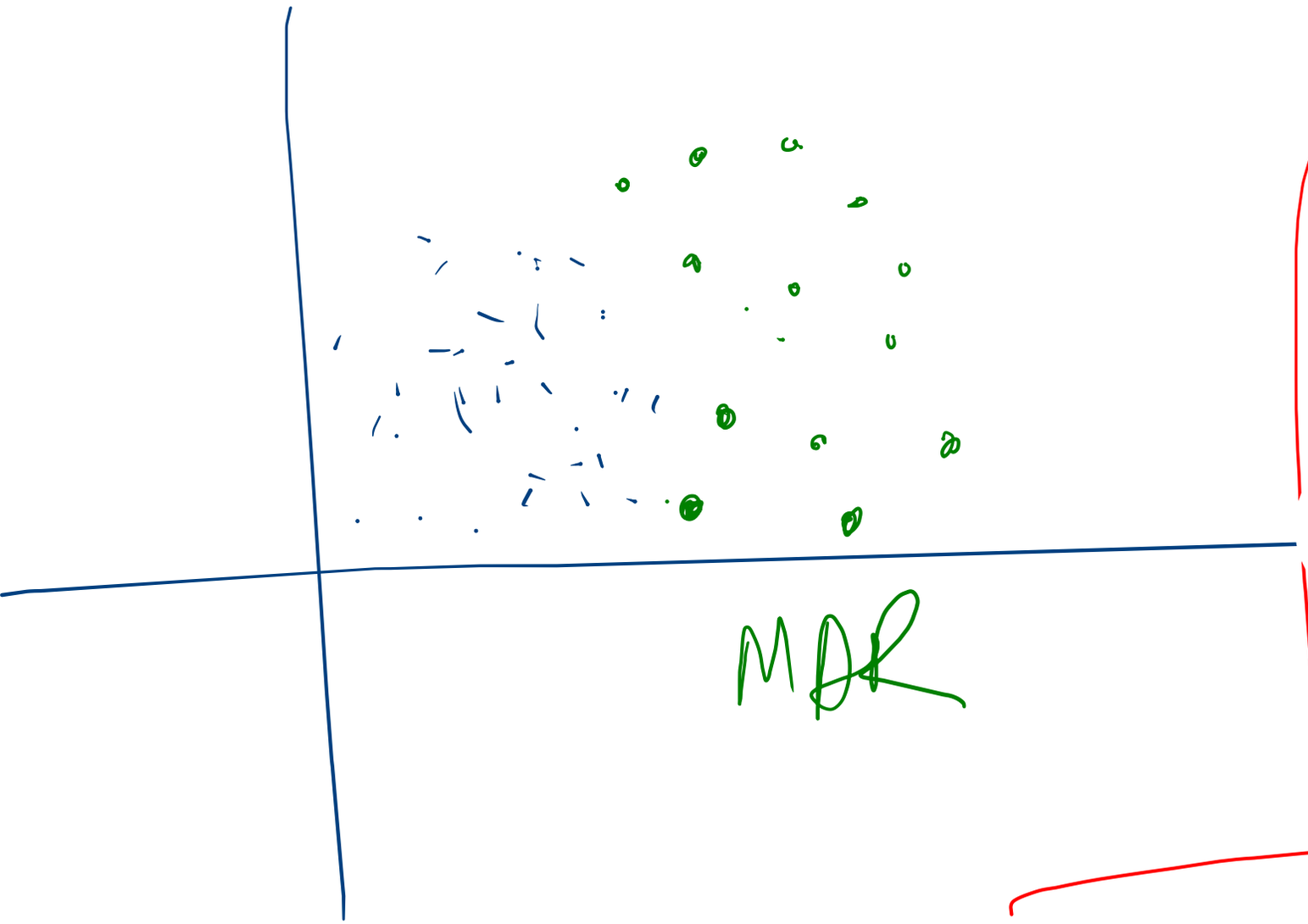
a

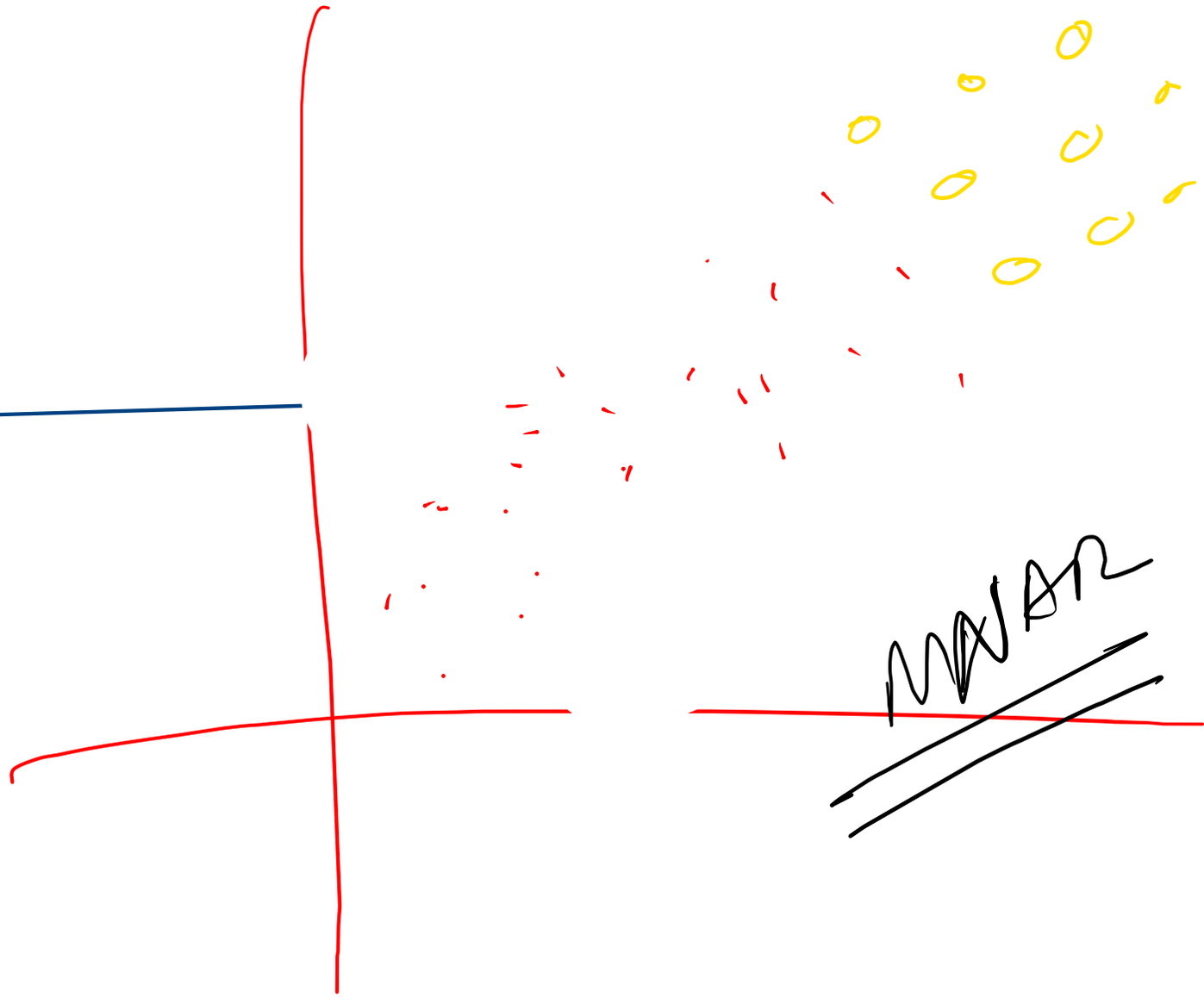
b

MDR

MCDR

MCDR


MDR


MDR

Detection

MCAR

Numerical

describe
isnull
isna

Visual

histogram
Scatterplot

obs. values

replace
ee missingno

Missingno Package

install missingno

import missingno as msno

bar

matrix

heatmap

dendogram

Y	R	Y_{mis}	Y_{obs}	X
0	1			
0	1			
.	.			
.	.			
—	0			
0	1			
—	0			
0	1			
—	1			
0	0			

$$R = \begin{cases} 0 & \text{if value missing} \\ 1 & \text{if value observed} \end{cases}$$

$$Y = (Y_{obs}, Y_{mis})$$

ψ : parameters

MD M ⇔ MCAR

$$P(R \mid Y_{obs}, Y_{mis}, \psi)$$

$R = 0$

Missing

$$P(R=0 \mid Y_{obs}, Y_{mis}, \psi) = P(R=0)$$

MAR

$$P\left(R=0 \mid Y_{obs}, Y_{mis}, \psi\right) = P\left(R=0 \mid Y_{obs}, \psi\right)$$

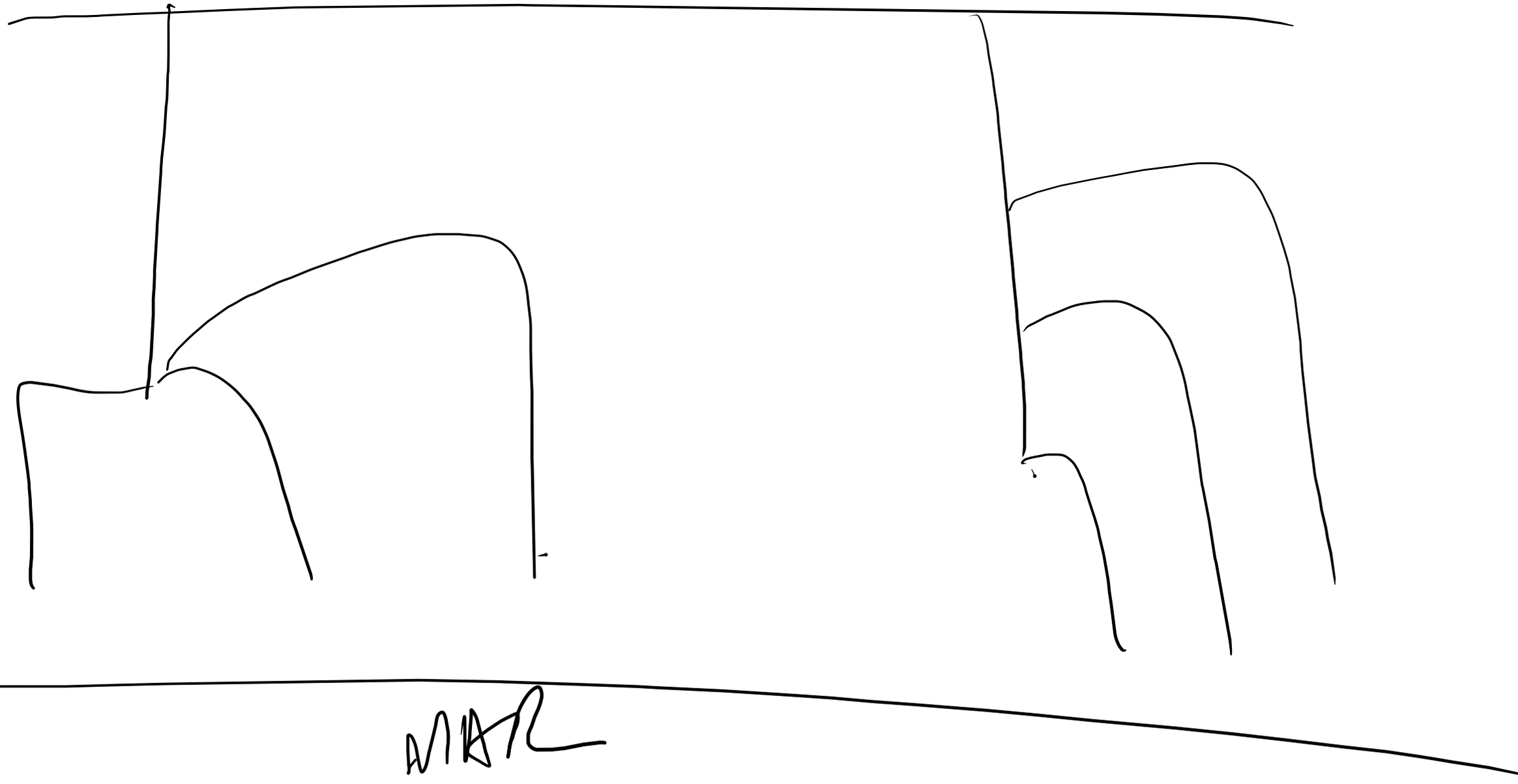
MNAR

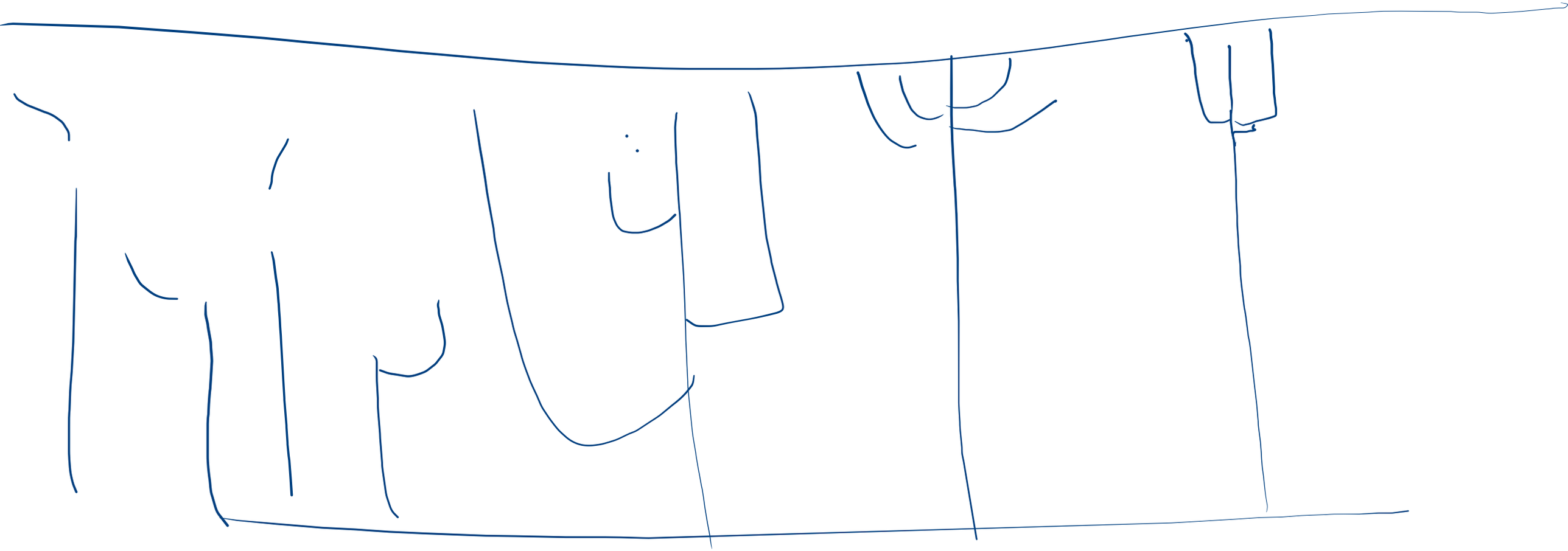
$$P\left(R=0 \mid Y_{obs}, Y_{mis}, \psi\right) =$$

MCAR

MNAR

MNAR







MSRO

bar

matrix

dendrogram

heat map

non-Confirmatory Signal

Statistical method / Test

Confirmatory tests

1. Little's MCAR Test (Overall test)
2. Dixon's MCAR Test (Univariate)
3. Hawkins MCAR Test (Multivariate)

Hypothesis tests

① Statement of the Test

H_0: Data MCAR

H_1: Not MCAR

Hypotheses → H_0 Null, H_1 Alternate

② α → Significance level

type of test → one-tailed, two-tailed

③ Test Statistic → χ^2

Compute the χ^2 test-statistic

Observed value χ^2 from the Sample data

→ p

Missing value Treatment

missingno.py →
- bar
- matrix
- heat map
- dendrogram

Little's MCAR test

Confirmatory tests

MCAR
- Ji Xan's
- Hawkin's test

H_0 : MCAR

H_1 : NOT MCAR

} α, two-tailed test

Reject

Test statistic

fast to reject

p

α 0.05

EDA

Univariate

Histogram

Q-Q plots

$\alpha = 0.05$, $\chi_0^2 = y$

s.t. $\chi_\alpha^2 = x$

$p = 0.26$

$\alpha = 0.05$

$p > \alpha \rightarrow$ Fail to Reject H_0

Missing Data MCAR

$p = 0.01$

$\alpha = 0.05$

$p < \alpha \rightarrow$ Reject H_0

Not MCAR

MCAR

Treatment

df.isnull

.isna

.info

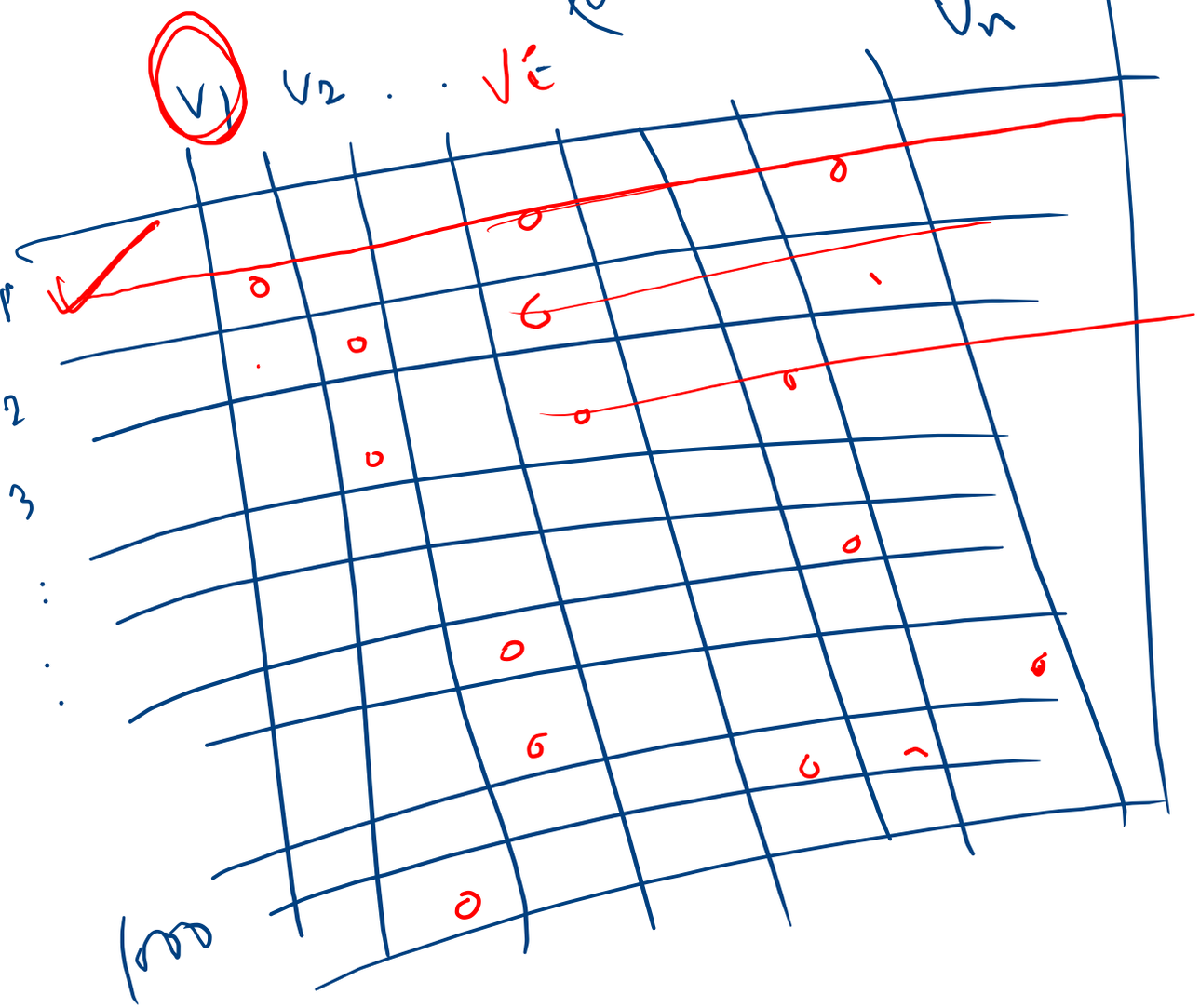
5 - 10 %

% missing values

Complete Case Analysis

Complete case (listwise deletion)

Available case (pairwise deletion)



Imputation

- Mean
- Median
- Function

} Numeric data

Regression

$$Y = C_0 + C_1 V_2 + C_3 V_3 + \varepsilon$$

Data transform

What is?

raw data → desired form

Why?

- reducing skewness
- removing outliers
- Comparing variables on a scale
- improve Model performance
- interpretability

When?

- before modeling
- Use any algorithm that needs assumption

How ?

Types

1 Scaling

- Z-score
- Min-max
- Robust

$$Z = \frac{X - \mu}{\sigma}$$

$$\mu = \frac{\sum x_i}{n}$$

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

$$S = \sqrt{\frac{\sum (x_i - \mu)}{n-1}}$$

$$\frac{x - x_{median}}{IQR}$$

$$\frac{x - x_{min}}{x_{max} - x_{min}}$$

$$E\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma} E(X-\mu)$$

$$= \frac{1}{\sigma}\left[E(X) - \mu\right]$$

$$= \frac{1}{\sigma}(\mu - \mu) = 0$$

Numeric

2

function

- log — highly right skewed, positive values
- square — Right Skewed
- Inverse — positive, Right skewed
- Square root —

Power

Box-Cox

$$y(\lambda) = \begin{cases} \dfrac{y^{\lambda}-1}{\lambda}, & \lambda \neq 0 \\ \log(y) & \text{if } \lambda = 0 \end{cases}$$

Yeo-Johnson

$$\psi(\lambda, y) = \begin{cases} \dfrac{(y+1)^{\lambda}-1}{\lambda} & \lambda \neq 0, y \geq 0 \\ \log(1+y) & \lambda = 0, y \geq 0 \\ -\log(1-y) & \lambda = 2, y = 0 \\ \dfrac{-(1-y)^{2-\lambda}-1}{2-\lambda}, & \lambda \neq 2, y = 0 \end{cases}$$

Quantile

Normal

$\text{rank}(\psi(x))$

Uniform

$y = \text{rank}(x)$

given data

Data Transformation

Stage 1 → features → values

- Compare
  - Scale
  - Shape

Numerical features

- Normalization
- Standardization
- Scaling (Robust) — sklearn
- function transf(x) = $\log x,\ x^2,\ \sqrt{x},\ x^{\frac{1}{n}}$
- power transf
  - Box-Cox
  - Yeo-Johnson
- quantile
  - Uniform — (Rank(x))
  - Normal — $\psi^{-1}(\text{Rank}(x))$ → Normal CDF

Stage II

Numeric ⟶ Category

→ Discretization / Binning

① Equal-width

② Equal-frequency

③ Domain-specific — Age — Infant, baby, ⋮

④ cluste-based — Group 1, Group 2, Group 3

[| | |] equal width



Stage III

Category → Numeric

Encoding

Gender

categories

M F T

1 0 0

0 1 0

0 0 1

- one-hot
- Label
  - 0
  - 1
  - 2
  - 3
  - ordinal variable
- ordinal
- Binary
  - 0 → 00
  - 1 → 01
  - 2 → 10
  - 3 → 11
- Target
  - mean
- Hash
  - entity